

Class test – 2 [MC3]

Q1.

The Hudson Engineering Group (HEG) has contacted you to create a conceptual model whose application will meet the expected database requirements of the company's training program. The description of the operating requirement is given below-

The HEG has 12 instructors and can handle up to 30 trainees per class. HEG offers five advanced technology courses each of which may generate several classes. If a class has fewer than 10 trainees, it will be cancelled. Therefore it is possible for a course not to generate any classes. Each class is taught by one instructor. Each instructor may teach up to two classes or may be assigned to do research only. Each trainee may take up to two classes per year. Draw the ER diagram for this situation.

- i Identify the entities. Take suitable attributes for each one of them.
- ii Identify relationships between/among entities.
- iii Specify the cardinality constraints, participation constraints and min, max constraint for each relation.
- iv Specify the assumptions made if any.

Write the SQL and relational Algebraic Query for all the case studies given below

Q2

Customer (CustID, email, Name, Phone, ReferrerID)

Bicycle (BicycleID, DatePurchased, Color, CustID, ModelNo)

BicycleModel (ModelNo, Manufacturer, Style)

Service (StartDate, BicycleID, EndDate)

- a) Identify primary and foreign keys. Create the tables and insert at least 5 records in each table.
- b) List all the customers who have the bicycles manufactured by manufacturer "Honda".
- c) List the bicycles purchased by the customers who have been referred by customer "C1".
- d) List the manufacturer of red colored bicycles.
- e) List the models of the bicycles given for service.

Class Test – 3 [MC3]

Q1. Consider a database with two objects X and Y and two transactions $T1$ and $T2$.

$T1$ performs the following operations: $R(X)$, $R(Y)$, $W(X)$. $T2$ performs the following operations: $R(X)$, $R(Y)$, $W(X)$, $W(Y)$.

- i) Give an example schedule with $T1$ and $T2$ that results in a write-read conflict.
- ii) Give an example schedule that results in a read-write conflict.
- iii) Give an example schedule that results in a write-write conflict.
- iv) For each of the three schedules, show that Strict 2PL disallows the schedule.

Q5. Perform the following operations in B Tree and B+ tree

The trees can hold up to 4 pointers and 3 keys

- a) Insert 1,3,5,7,9,2,4,6,8,10
- b) Delete 9,7,8